| | |
|---|---|
| United States Patent | 12416441 |
| Kind Code | B2 |
| Date of Patent | September 16, 2025 |
| Inventor(s) | Shirata; Mitsuyuki et al. |


## Cold storage


### Abstract

A cold storage includes a box including a cold storage chamber, a sliding door that opens and closes the cold storage chamber, and a machine chamber; a compressor disposed in the machine chamber and constituting a refrigeration circuit that cools the inside of the cold storage chamber; a fan that generates an airflow passing around the compressor and blown to the sliding door through an air outlet opened to the machine chamber; a heater disposed in the cold storage chamber; and a control apparatus that operates the heater when a condition of a set temperature for the cold storage chamber being equal to or lower than a specified temperature and a condition of the compressor being in an operating state are satisfied.



**Inventors:** **Shirata; Mitsuyuki (Ehime, JP), Yanagihara; Naoki (Gunma, JP)**

**Applicant:** **PHC Holdings Corporation** (Tokyo, JP)



**Family ID:** **1000008821569**

**Assignee:** **PHC Holdings Corporation (Tokyo, JP)**

**Appl. No.:** **18/365733**

**Filed:** **August 04, 2023**

### Prior Publication Data

| Document Identifier | Publication Date |
|---|---|
| US 20240019192 A1 | Jan. 18, 2024 |

### Foreign Application Priority Data

| | | |
|---|---|---|
| JP | 2021-043838 | Mar. 17, 2021 |

### Related U.S. Application Data

continuation parent-doc WO PCT/JP2022/004485 20220204 PENDING child-doc US 18365733

## Publication Classification

**Int. Cl.:** **F25D21/08** (20060101); **F25B13/00** (20060101)

**U.S. Cl.:**

CPC **F25D21/08** (20130101); **F25B13/00** (20130101);

## Field of Classification Search

**CPC:** A47F (3/0426); F25B (13/00); F25D (21/006); F25D (21/08); F25D (23/006); F25D (23/021); F25D (2700/12)

## References Cited

**U.S. PATENT DOCUMENTS**

| Patent No. | Issued Date | Patentee Name | U.S. Cl. | CPC |
|---|---|---|---|---|
| 4977754 | 12/1989 | Upton | 62/277 | A47F 3/0408 |
| 6550261 | 12/2002 | Shima | 219/385 | F25D 21/04 |
| 2011/0214444 | 12/2010 | Yoshii | 62/160 | F25B 47/006 |
| 2022/0170687 | 12/2021 | Tsukui et al. | N/A | N/A |

**FOREIGN PATENT DOCUMENTS**

| Patent No. | Application Date | Country | CPC |
|---|---|---|---|
| 4743299 | 12/1999 | AU | N/A |
| 02-277411 | 12/1989 | JP | N/A |
| 2000-088438 | 12/1999 | JP | N/A |
| 2001-174119 | 12/2000 | JP | N/A |
| 3190793 | 12/2000 | JP | N/A |
| 2001-245759 | 12/2000 | JP | N/A |
| 2001-174119 | 12/2000 | JP | N/A |
| 2005-164199 | 12/2004 | JP | N/A |
| 2017-020745 | 12/2016 | JP | N/A |
| 2021/039446 | 12/2020 | WO | N/A |

**OTHER PUBLICATIONS**

European Office Action dated Jan. 13, 2025 issued in the corresponding European Patent Application No. 22770930.0. cited by applicant
Extended European Search Report dated Apr. 30, 2024 from corresponding EP Application No. 22770930.0, 8 pages. cited by applicant
International Search Report dated Mar. 8, 2022 issued in International Patent Application No. PCT/JP2022/004485, with English translation. cited by applicant

*Primary Examiner:* Crenshaw; Henry T

*Attorney, Agent or Firm:* Rimon P.C.

## Background/Summary

CROSS-REFERENCE OF RELATED APPLICATIONS (1) This application is a Continuation of International Patent Application No. PCT/JP2022/004485, filed on Feb. 4, 2022, which in turn claims the benefit of Japanese Patent Application No. 2021-043838, filed on Mar. 17, 2021, the entire disclosures of which Applications are incorporated by reference herein.

TECHNICAL FIELD
(1) The present disclosure relates to a cold storage.
BACKGROUND ART
(2) Patent Literature (PTL) 1 discloses a cooling storage cabinet in which dew condensation on a glass door can be effectively eliminated by utilizing air from a condenser fan.
CITATION LIST
Patent Literature
(3) PTL 1 Japanese Patent Application Laid-Open No. 2000-88438
SUMMARY OF INVENTION
Technical Problem
(4) A cold storage stores pharmaceuticals and so on in a cold storage chamber at a low temperature. Dew condensation generates on a sliding door for opening and closing the cold storage chamber due to a temperature difference between the inside and the outside of the cold storage chamber. There is a possibility that an amount of the dew condensation generated on the sliding door may increase as the temperature in the cold storage chamber becomes lower. Furthermore, the cold storage chamber has been designed to be cooled to an even lower temperature in recent years. Such a trend may lead to a possibility of causing a larger amount of the dew condensation to generate on the sliding door.
(5) An object of the present disclosure is to suppress the generation of the dew condensation in the cold storage.
Solution to Problem
(6) In order to achieve the above mentioned object, a cold storage according to the present disclosure includes: a box including a cold storage chamber, a sliding door that opens and closes the cold storage chamber, and a machine chamber; a compressor disposed in the machine chamber and constituting a refrigeration circuit that cools an inside of the cold storage chamber; a fan that generates an airflow passing around the compressor and blown to the sliding door through an air outlet opened to the machine chamber; a heater disposed in the cold storage chamber; and a control apparatus that operates the heater when a condition of a set temperature for the cold storage chamber being equal to or lower than a specified temperature and a condition of the compressor being in an operating state are satisfied.
Advantageous Effects of Invention
(7) With the cold storage according to the present disclosure, the generation of the dew condensation can be suppressed.

---

## Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a perspective view of a cold storage according to an embodiment of the present disclosure;
(2) FIG. **2** is a horizontal sectional view illustrating a configuration inside a machine chamber;
(3) FIG. **3** is a partial vertical sectional view of the cold storage;
(4) FIG. **4** is a vertical sectional view illustrating a cooling region inside a cold storage chamber;

(5) FIG. **5** is a block diagram of the cold storage;
(6) FIG. **6** is a flowchart of a program executed by a control apparatus;
(7) FIG. **7** is a flowchart of a program executed by the control apparatus;
(8) FIG. **8** is a flowchart of a program executed by the control apparatus; and
(9) FIG. **9** is a time chart illustrating an operation of a drain pan.

DESCRIPTION OF EMBODIMENTS

(10) An embodiment of a cold storage according to the present disclosure will be described below with reference to the drawings. In the following description, it is assumed that, as denoted by arrows in FIG. **1**, a side where sliding door **30** is disposed is a front side with respect to cold storage **1**, and a side opposite to the above side is a rear side with respect to cold storage **1**. Left and right sides when cold storage **1** is viewed from the front are respectively left and right sides with respect to cold storage **1**. A side away from a plane on which cold storage **1** is installed is an upper side with respect to cold storage **1**, and a side opposite to the above side is a lower side with respect to cold storage **1**.

(11) Cold storage **1** is a pharmaceutical cold storage for storing pharmaceuticals at a low temperature. Cold storage **1** may be a cold storage for blood or a constant temperature container. As illustrated in FIGS. **1** to **4**, cold storage **1** includes box **10**, frame **20**, and sliding door **30**.

(12) Box **10** has, in its front surface, opening H**1** that provides an entrance opened and closed with movement of sliding door **30**. A heat insulating material is filled between an outer surface and an inner surface of box **10**. A space surrounded by the inner surface of box **10** serves as cold storage chamber R**1**, namely a space in which pharmaceuticals are stored (FIGS. **3** and **4**).

(13) Frame **20** is attached to box **10** in such a way of bordering opening H**1**. Frame **20** includes outer rail **21** and inner rail **22** that are disposed on a bottom surface of frame **20** and that extend in a left-right direction. Inner rail **22** is positioned rearward of outer rail **21** (on a side nearer to cold storage chamber R**1**) (FIG. **3**). In addition, sliding door **30** is attached to frame **20**.

(14) Sliding door **30** includes first sliding door **31** and second sliding door **32**. First sliding door **31** is attached to be movable along outer rail **21**. First sliding door **31** is positioned in a right-side region within frame **20** in a closed state. Second sliding door **32** is attached to be movable along inner rail **22**. Second sliding door **32** is positioned in a left-side region within frame **20** in a closed state. Opening H**1** and hence cold storage chamber R**1** are opened and closed with movements of first sliding door **31** and second sliding door **32**.

(15) Since first sliding door **31** is disposed on outer rail **21** and second sliding door **32** is disposed on inner rail **22**, second sliding door **32** is positioned on a side nearer to cold storage chamber R**1** than first sliding door **31**. Accordingly, an amount of dew condensation generated on second sliding door **32** is larger than that generated on first sliding door **31**.

(16) Box **10** further includes machine chamber R**2** under cold storage chamber R**1** (FIGS. **2** and **3**).

(17) Machine chamber R**2** has air outlet H**2**. Air outlet H**2** is formed to be opened in front of frame **20** such that an airflow (described later) is blown to sliding door **30**.

(18) Compressor **41** and condenser **42**, both constituting a refrigeration circuit for cooling the inside of cold storage chamber R**1**, and fan **43** are disposed in machine chamber R**2**. Compressor **41** is disposed in a left-side region within machine chamber R**2**. In other words, compressor **41** is disposed at a position nearer to second sliding door **32** in a state of closing cold storage chamber R**1** than first sliding door **31** in a state of closing cold storage chamber R**1**. Condenser **42** is disposed in machine chamber R**2** substantially at a center.

(19) Fan **43** generates an airflow. The airflow is denoted by thick arrows in FIGS. **1** and **3**. Fan **43** is rotated to take outside air into machine chamber R**2** and to generate the airflow. The airflow passes around compressor **41** and condenser **42**. When compressor **41** and condenser **42** are in an operating state, the airflow passing around compressor **41** and condenser **42** is warmed by the temperature of compressor **41** and the temperature of condenser **42**.

(20) Furthermore, the airflow is blown out through air outlet H**2** and is sprayed to sliding door **30**.

Because the warmed airflow warms sliding door **30**, generation of the dew condensation on sliding door **30** is suppressed.
(21) As illustrated in FIG. **4**, cold storage chamber R**1** is partitioned by sidewall **51** into storage region R**1***a* and cooling region R**1***b*. Storage region R**1***a* is a region where the pharmaceuticals and so on are stored. Cooling region R**1***b* is a region where the air in cold storage chamber R**1** is cooled.
(22) Second fan **52**, evaporator **53** constituting the refrigeration circuit, sensor **54**, defrost heater **55**, second sensor **56**, drain pan **57**, and drain pan heater **58** are disposed in an upper end portion of cold storage chamber R**1** on the rear side. The surrounding of evaporator **53** serves as cooling region R**1***b*. Stated another way, second fan **52**, evaporator **53** constituting the refrigeration circuit, sensor **54**, defrost heater **55**, second sensor **56**, drain pan **57**, and drain pan heater **58** are disposed in cooling region R**1***b*.
(23) Second fan **52** is rotated to take air in storage region R**1***a* into cooling region R**1***b*. Second fan **52** is disposed in an upper end zone of cooling region R**1***b*. Accordingly, second fan **52** takes in air present in an upper zone of storage region R**1***a*. The air having been taken into cooling region R**1***b* is blown into storage region R**1***a* through an opening that is formed at the bottom of cooling region R**1***b*. Thus, as indicated by an arrow in FIG. **4**, the air having been taken into cooling region R**1***b* flows downward from the upper end zone of cooling region R**1***b*.
(24) Evaporator **53** cools the air having been taken into cooling region R**1***b*. Evaporator **53** is disposed on a lower side than second fan **52**. Evaporator **53** includes pipe **53***a* through which a coolant circulating in the refrigeration circuit flows, and fin **53***b* attached to be held in contact with pipe **53***a*.
(25) Sensor **54** detects a temperature in cold storage chamber R**1**. Sensor **54** is disposed in cooling region R**1***b* on an upper side than evaporator **53**. In other words, sensor **54** detects a temperature of the air taken into cooling region R**1***b* before the taken-in air is cooled by evaporator **53**. Thus, the temperature detected by sensor **54** is equal to the temperature in the air in storage region R**1***a*.
(26) Defrost heater **55** serves as a heater for, when operated, melting frost adhering to pipe **53***a* and fin **53***b*. Defrost heater **55** is, for example, a sheath heater or a cord heater. Defrost heater **55** is attached at a position away from pipe **53***a* of evaporator **53** while it is held in contact with fin **53***b*. An operation started with activation of defrost heater **55** is especially referred to as a “defrost operation”. The defrost operation is performed during a period in which compressor **41** is stopped.
(27) Second sensor **56** is disposed at a position away from pipe **53***a* while it is held in contact with fin **53***b*. Second sensor **56** detects a temperature of fin **53***b*.
(28) Drain pan **57** receives water generated due to the defrost operation. Drain pan **57** is disposed under evaporator **53**. The frost adhering to pipe **53***a* and fin **53***b* is melted by the defrost operation, whereby water is generated. The generated water falls down onto drain pan **57** and is drained to machine chamber R**2** through a pipe (not illustrated).
(29) Drain pan heater **58** is a heater for heating drain pan **57**. Drain pan heater **58** is, for example, a sheath heater or a cord heater. An amount of heat generated by drain pan heater **58** is smaller than that generated by defrost heater **55**. Drain pan heater **58** is attached to be held in contact with a rear surface of drain pan **57**. There is a possibility that the water received by drain pan **57** may be frozen by being cooled by evaporator **53**. Even when the water received by drain pan **57** is frozen and an ice is generated, the ice can be melted with the operation of drain pan heater **58**.
(30) Defrost heater **55** and drain pan heater **58** are disposed in cooling region R**1***b* as described above. In other words, defrost heater **55** and drain pan heater **58** are disposed in cold storage chamber R**1**.
(31) As illustrated in FIG. **5**, cold storage **1** further includes inputter **61** and control apparatus **62**. Inputter **61** is used to input a set temperature for cold storage chamber R**1**. Inputter **61** is, for example, a touch panel.
(32) Control apparatus **62** is a computer for supervising and controlling cold storage **1**. Control

apparatus **62** includes a storage apparatus for storing a computer program (hereinafter simply referred to as a “program”) and a processor for executing the computer program.

(33) Inputter **61**, sensor **54**, second sensor **56**, compressor **41**, defrost heater **55**, drain pan heater **58**, fan **43**, and second fan **52** are electrically connected to control apparatus **62**. Control apparatus **62** obtains the set temperature input through inputter **61**, the temperature detected by sensor **54**, and the temperature detected by second sensor **56**. Control apparatus **62** controls compressor **41**, defrost heater **55**, drain pan heater **58**, fan **43**, and second fan **52** based on the set temperature, the temperature detected by sensor **54**, and the temperature detected by second sensor **56**.

(34) With control apparatus **62** executing the program, compressor **41** is controlled, and hence the temperature in cold storage chamber R**1** is controlled to the set temperature. In more detail, control apparatus **62** repeats operation and stop of compressor **41** based on the set temperature and the temperature detected by sensor **54** (FIG. **9**). When the program is executed, fan **43** and second fan **52** are controlled to rotate continuously.

(35) The operation of drain pan heater **58**, realized with control apparatus **62** executing the program, will be described below with reference to flowcharts of FIGS. **6** to **8**. At the start time of the program, drain pan heater **58** is not operated. Furthermore, when the program is executed, the operation and the stop of Compressor **41** are repeated.

(36) In S**10** illustrated in FIG. **6**, control apparatus **62** determines whether or not the set temperature obtained from inputter **61** is equal to or lower than a specified temperature. The specified temperature is given as, for example, 3° C., taking into consideration that the amount of the dew condensation generated on sliding door **30** increases relatively when the temperature in cold storage chamber R**1** reaches the specified temperature. The specified temperature is previously set in the program executed by control apparatus **62** in a manufacturing stage of cold storage **1**. If the set temperature is equal to or lower than the specified temperature (S**10**: YES), control apparatus **62** executes synchronous control (described later) in S**11**.

(37) On the other hand, if the set temperature is higher than the specified temperature (S**10**: NO), control apparatus **62** executes asynchronous control (described later) in S**12**.

(38) The case in which the synchronous control, illustrated in FIG. **7**, is executed will be described below. In the synchronous control, drain pan heater **58** is operated when compressor **41** is operated.

(39) In S**20**, control apparatus **62** determines whether or not compressor **41** is in the operating state. If compressor **41** is stopped (S**20**: NO), control apparatus **62** repeatedly executes S**20**.

(40) On the other hand, if compressor **41** is in the operating state (S**20**: YES), control apparatus **62** determines in S**21** whether or not the temperature detected by sensor **54** is equal to or lower than a temperature (hereinafter referred to as a “sum temperature”) resulting from adding a specified value to the set temperature. The specified value is a value at which, regardless of the set temperature being any temperature, the amount of the dew condensation generated on sliding door **30** decreases relatively when the temperature in cold storage chamber R**1** reaches the sum temperature. The specified value is a constant value irrespective of the set temperature and is previously set in the program executed by control apparatus **62** in the manufacturing stage of cold storage **1**. The specified value is, for example, 5.

(41) If the temperature in cold storage chamber R**1** is relatively higher than the set temperature and hence the temperature detected by sensor **54** is higher than the sum temperature (S**21**: NO), control apparatus **62** returns the program to S**20** without operating drain pan heater **58**.

(42) On the other hand, if, because of compressor **41** being in the operating state, the temperature in cold storage chamber R**1** drops and the temperature detected by sensor **54** becomes equal to or lower than the sum temperature (S**21**: YES), control apparatus **62** operates drain pan heater **58** in S**22**.

(43) Then, control apparatus **62** determines in S**23** whether or not compressor **41** is stopped. If the temperature in cold storage chamber R**1** does not reach the set temperature and hence compressor **41** remains in the operating state (S**23**: NO), control apparatus **62** repeatedly executes S**23**.

(44) On the other hand, if the temperature in cold storage chamber R**1** reaches the set temperature and hence compressor **41** is stopped (S**23**: YES), control apparatus **62** stops drain pan heater **58** in S**24** and returns the program to S**20**.

(45) The case in which the asynchronous control, illustrated in FIG. **8**, is executed will be described below. In the asynchronous control, the operation of compressor **41** and the operation of drain pan heater **58** are not in synchronism with each other.

(46) In S**30**, control apparatus **62** determines whether or not the operating state of compressor **41** is stopped. If the operating state of compressor **41** is continued (S**30**: NO), control apparatus **62** repeatedly executes S**30**.

(47) On the other hand, if the temperature in cold storage **1** reaches the set temperature and hence the operation of compressor **41** is stopped (S**30**: YES), control apparatus **62** operates drain pan heater **58** and starts the defrost operation, more specifically, the operation of defrost heater **55** in S**31**. The defrost operation is continued until the temperature detected by second sensor **56** reaches a second specified temperature.

(48) Then, control apparatus **62** determines in S**32** whether or not the temperature detected by second sensor **56** is equal to or higher than a third specified temperature. The third specified temperature is a threshold set to stop the operation of drain pan heater **58**. The third specified temperature is higher than the second specified temperature. The second specified temperature and the third specified temperature are previously set in the program executed by control apparatus **62** in the manufacturing stage of cold storage **1**.

(49) If the temperature detected by second sensor **56** is lower than the third specified temperature (S**32**: NO), control apparatus **62** repeatedly executes S**32**.

(50) If the temperature detected by second sensor **56** becomes equal to or higher than the third specified temperature (S**32**: YES), control apparatus **62** stops drain pan heater **58** in S**33** and returns the program to S**30**.

(51) The temperature in cold storage chamber R**1** when the temperature detected by second sensor **56** becomes the third specified temperature is at a level that is sufficiently lower than a temperature at which there is a possibility of adversely affecting the pharmaceuticals and so on stored in cold storage chamber R**1**.

(52) The operation of cold storage **1** when the synchronous control of drain pan heater **58** is performed will be described below with reference to a time chart of FIG. **9**. When the program is executed, fan **43** is continuously rotated, and hence the airflow is blown to sliding door **30** as described above.

(53) If the set temperature is equal to or lower than the specified temperature (S**10**: YES) and if the temperature detected by sensor **54** is equal to or lower than the sum temperature (S**21**: YES) at the time when compressor **41** starts the operation (S**20**: YES) (time t**1**), drain pan heater **58** is operated (S**22**).

(54) Because compressor **41** is operated, the air in cold storage chamber R**1** is cooled by evaporator **53**. On the other hand, because drain pan heater **58** is disposed in cold storage chamber R**1**, drain pan heater **58** heats the air cooled by evaporator **53**. Thus, drain pan heater **58** works to suppress the cooling performed by the refrigeration circuit. Accordingly, when drain pan heater **58** is operated, a time taken for the temperature in cold storage chamber R**1** to reach the set temperature is longer than that when drain pan heater **58** is not operated. In other words, an operating time of compressor **41** when drain pan heater **58** is operated is longer than that when drain pan heater **58** is stopped.

(55) As the operating time of compressor **41** increases, the temperature of compressor **41** rises. Due to the rise of the compressor temperature, the temperature of the airflow passing around compressor **41** also rises. In other words, the temperature of the airflow when drain pan heater **58** is operated becomes higher than that when drain pan heater **58** is stopped. Thus, when the set temperature is set to be equal to or lower than the specified temperature at which the amount of the

dew condensation generated on sliding door **30** increases relatively, the temperature of the airflow can be raised, and hence the generation of the dew condensation can be suppressed.

(56) As described above, compressor **41** is disposed nearer to second sliding door **32** in the state of closing cold storage chamber R**1** than to first sliding door **31** in the state of closing cold storage chamber R**1**. Accordingly, the temperature of the airflow blown to second sliding door **32** can be made higher than that blown to first sliding door **31**. Moreover, as described above, the amount of the dew condensation generated on second sliding door **32** is larger than that generated on first sliding door **31**. In other words, the amount of the dew condensation generated on second sliding door **32** can be effectively suppressed depending on the layout position of compressor **41**.

(57) When compressor **41** is stopped upon the temperature in cold storage chamber R**1** dropping and reaching the set temperature (S**23**: YES, time t**2**), drain pan heater **58** is stopped (S**24**).

(58) Then, if the set temperature is equal to or lower than the specified temperature (S**10**: YES) and if the temperature detected by sensor **54** is higher than the sum temperature (S**21**: NO) even at the time when compressor **41** starts the operation (S**20**: YES) (time t**4**), drain pan heater **58** is not operated. When the temperature detected by sensor **54** is higher than the sum temperature, the difference between the temperature in cold storage chamber R**1** and the set temperature is relatively large. Therefore, drain pan heater **58** is controlled to be not operated with higher priority given to the cooling by the refrigeration circuit.

(59) The case in which the temperature detected by sensor **54** is higher than the sum temperature corresponds to, for example, the case in which the outside air flows into cold storage chamber R**1** because of sliding door **30** being opened by a user and the temperature in cold storage chamber R**1** rises. Moreover, when the temperature detected by sensor **54** is higher than the sum temperature, the amount of the dew condensation generated on sliding door **30** is relatively small as described above. Accordingly, the generation of the dew condensation can be sufficiently suppressed with the airflow even at the temperature when drain pan heater **58** is not operated.

(60) Then, when the temperature in cold storage chamber R**1** drops due to the operation of compressor **41** and the temperature detected by sensor **54** becomes equal to or lower than the sum temperature (S**21**: YES), drain pan heater **58** is operated (S**22**, time t**5**). Moreover, when compressor **41** is stopped upon the temperature in cold storage chamber R**1** further dropping and reaching the set temperature (S**23**: YES, time t**6**), drain pan heater **58** is stopped (S**24**).

(61) The operation of cold storage **1** when the asynchronous control of drain pan heater **58** is performed will be described below with reference to the time chart of FIG. **9**. When the asynchronous control is performed, fan **43** is continuously rotated, and the airflow is blown to sliding door **30** as in the above-described case.

(62) In the state of the set temperature being higher than the specified temperature (S**10**: NO), at the time when compressor **41** starts the operation (S**30**: NO) (time t**1**), drain pan heater **58** is not operated. Accordingly, the temperature of the airflow blown to sliding door becomes lower than that when drain pan heater **58** is operated. However, when the set temperature is higher than the specified temperature, the amount of the dew condensation generated on sliding door **30** is smaller than that generated on sliding door **30** when the set temperature is equal to or lower than the specified temperature. As a result, the dew condensation generated on sliding door **30** can be sufficiently suppressed.

(63) When the operation of compressor **41** is stopped upon the temperature in cold storage chamber R**1** dropping and reaching the set temperature (S**30**: YES, time t**2**), drain pan heater **58** is operated (S**31**). At that time (t**2**), defrost heater **55** is also operated.

(64) Because compressor **41** is stopped, the temperature of compressor **41** does not rise in spite of drain pan heater **58** being operated. Accordingly, the temperature of the airflow does not rise.

(65) When the temperature detected by second sensor **56** becomes equal to or higher than the third specified temperature due to melting of the frost adhering to fin **53***b* and rising of the temperature of fin **53***b* (S**32**: YES, time t**3**), the operation of drain pan heater **58** is stopped (S**33**).

(66) Then, drain pan heater **58** repeatedly executes processes of starting the operation when the operation of compressor **41** is stopped (time t**6**) and stopping the operation when the temperature detected by second sensor **56** becomes equal to or higher than the third specified temperature (t**7**).
(67) The present disclosure is not limited to the above-described embodiment. Modifications obtained by variously modifying the embodiment also fall within the scope of the present disclosure insofar as the modifications do not depart from the gist of the present disclosure.
(68) For example, in the synchronous control, when compressor **41** is in the operating state, drain pan heater **58** may be operated irrespective of the sum temperature. In that case, S**21** in the flowchart illustrated in FIG. **7** is not executed. Stated another way, control apparatus **62** operates drain pan heater **58** when a condition of the set temperature in cold storage chamber R**1** being equal to or lower than the specified temperature and a condition of compressor **41** being in the operating state are satisfied.
(69) Furthermore, compressor **41** may be disposed at the center of machine chamber R**2** or at a position nearer to first sliding door **31** in the state of closing cold storage chamber R**1** than second sliding door **32** in the state of closing cold storage chamber R**1**.
(70) Moreover, in the synchronous control, defrost heater **55** may be operated instead of drain pan heater **58** when compressor **41** is in the operating state.
(71) The disclosure of Japanese Patent Application No. 2021-043838 filed on Mar. 17, 2021 including the specification, claims, drawings and abstract is incorporated herein by reference in its entirety.
INDUSTRIAL APPLICABILITY
(72) The present disclosure can be widely applied to cold storages, such as a cold storage for pharmaceuticals, a cold storage for blood, and a constant temperature container.
REFERENCE SIGNS LIST
(73) **1** cold storage **10** box **20** frame **30** sliding door **41** compressor **42** condenser **43** fan **52** second fan **54** sensor **56** second sensor **57** drain pan **58** drain pan heater H**1** opening H**2** air outlet R**1** cold storage chamber R**2** machine chamber

## Claims

1. A cold storage comprising: a box including a cold storage chamber, a sliding door that opens and closes the cold storage chamber, and a machine chamber; a compressor disposed in the machine chamber and constituting a refrigeration circuit that cools an inside of the cold storage chamber; a fan that generates an airflow passing around the compressor and blown to the sliding door through an air outlet opened to the machine chamber; a heater disposed in the cold storage chamber; a sensor that detects a temperature in the cold storage chamber; one or more memories; and at least one processor each coupled to at least one of the one or more memories and configured to operate the heater when a condition of a set temperature for the cold storage chamber being equal to or lower than a specified temperature, a condition of the compressor being in an operating state, and a temperature detected by the sensor being equal to or lower than a temperature resulting from adding a specified positive value to the set temperature are satisfied.
2. The cold storage according to claim 1, wherein: the sliding door includes a first sliding door that opens and closes the cold storage chamber, and a second sliding door that opens and closes the cold storage chamber at a position between the cold storage chamber and the first sliding door, and the compressor is disposed at a position nearer to the second sliding door in a state of closing the cold storage chamber than the first sliding door in a state of closing the cold storage chamber.
3. The cold storage according to claim 2, further comprising a drain pan that receives water generated due to a defrost operation, wherein the heater is a drain pan heater that heats the drain pan.
4. The cold storage according to claim 3, further comprising: a second fan; and an evaporator,

wherein the second fan generates a second airflow sequentially passing through the evaporator, the drain pan, and the drain pan heater.

5. The cold storage according to claim 4, wherein: the cold storage chamber is partitioned by a sidewall into a storage region and a cooling region, the storage region has an opening, the second fan, the evaporator, the drain pan, and the drain pan heater are located inside the cooling region, and the second airflow further sequentially passing through the opening and the cooling region.

6. The cold storage according to claim 5, wherein the cooling region is located above the machine chamber.